

# GDP Growth and Unemployment in the Netherlands and Western Europe (2015–2025)

Jakob, Ben, and David

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## Set-up your environment

## GDP Growth and Unemployment in the Netherlands and Western Europe

**Authors:** Jakob Vahtera Persson, Ben Akyureklier, David Craveiro

**Tutorial group:** 2\_5

**Tutorial lecturer:** F. Amankour (Fatima)

## Part 1 - Identify a Social Problem

### 1.1 Describe the Social Problem

Unemployment is one of the most closely watched indicators of economic well-being, because joblessness carries costs that go beyond lost income: it is associated with poorer mental and physical health, lower lifetime earnings, and weaker social cohesion (Eurofound, 2017). A central question for policymakers is how unemployment responds to the wider economy. Okun’s Law describes a stable negative relationship between output growth and unemployment (Okun, 1962): when GDP grows, unemployment tends to fall, and vice versa.

Understanding the strength of this link matters for policy. If unemployment reacts sharply to downturns, governments may need stronger automatic stabilisers; if it reacts weakly — as during the 2020 COVID-19 shock, when wage-subsidy schemes kept workers employed — the usual relationship can break down. This report quantifies the relationship between GDP growth and unemployment in the Netherlands (2015-2025), compares unemployment across Western Europe, and examines how youth workers are disproportionately exposed.

Existing work establishes Okun’s Law as a broad empirical regularity, but two gaps remain underexplored. First, the coefficient is not stable across countries: estimates for the Netherlands differ markedly from euro-area averages, yet country-specific deviations are often glossed over in cross-country studies (Ball et al., 2017). Second, the COVID-19 period may represent a structural break rather than a normal cyclical episode, since large-scale job-retention schemes severed the usual output–unemployment link in ways the pre-2020 literature could not anticipate. This report addresses both by focusing on a single country and treating 2020 as a potential break rather than a data point on a stable line.

## Part 2 - Data Sourcing

### 2.1 Load in the data

We use three datasets downloaded from Eurostat, the statistical office of the European Union. The exact tables are:

- Real GDP growth rate: Eurostat table `tec00115` (annual % change).
- Total unemployment rate: Eurostat table `tesem120`.
- Youth unemployment rate (ages 15-24): Eurostat table `tips1m80`.

The raw extracts are stored in the `data/` folder and loaded in the pipeline at the top of this document.

We rely on Eurostat rather than CBS or the OECD because the spatial comparison in Section 3.4 only works if every country's unemployment rate is built the same way. CBS would in principle give a more precise series for the Netherlands, since it is the original data collector and Eurostat's Dutch figures are themselves derived from CBS submissions, but it has no equivalent for Spain or Italy, so a CBS-based analysis could not have supported the cross-country part of this report at all. The OECD was considered too, but its unemployment definitions are not always applied identically across member states, which would have undermined the same comparability the spatial map depends on. Going through Eurostat rather than CBS directly means our Dutch figures pass through an extra harmonisation step we cannot inspect, and any rounding or adjustment Eurostat applies before publishing is invisible to us. We also take Eurostat's definition of the labour force as given, rather than checking it against how CBS itself frames unemployment domestically. The Dutch trend we report matches widely cited figures for the period, so we don't think this changes our conclusions, but the spatial and time-series parts of the analysis both depend on the same Eurostat methodology, which is also why the definition-mismatch limitation in Section 4.1 cannot be fixed simply by picking a different source.

### 2.2 Provide a short summary of the dataset(s)

Each dataset reports an annual percentage value per country. The unemployment series express the number of unemployed as a share of the active labour force; the GDP series expresses the annual percentage change in real GDP. All values are sourced from Eurostat and cover the period 2015-2025.

### 2.3 Describe the type of variables included

The variables are socio-economic indicators. Unemployment figures come from the EU Labour Force Survey, a large harmonised household survey, while GDP figures come from national accounts compiled by national statistical institutes. The data is therefore official administrative/survey data, comparable across countries, rather than self-reported by individuals.

## Part 3 - Quantifying

### 3.1 Data cleaning

All cleaning steps were run in the data pipeline at the top of this file to keep the workflow reproducible. For each dataset we (1) filtered for the Netherlands, (2) kept only the year and value columns, (3) recoded the placeholder -99 as NA and dropped missing values, and (4) joined the three series by year.

Two of these decisions are worth questioning. Dropping rows where the value is missing is the simplest option, but for such a short annual series it discards information; multiple imputation or carrying the series

forward could preserve sample size, at the cost of inventing values we cannot verify. More importantly, recoding the placeholder `-99` as NA throws away the reason a value was missing: Eurostat uses distinct flags for “not available”, “confidential”, and “low reliability”, and collapsing them into a single NA loses that signal. A refinement would be to retain the original status flags and decide case-by-case, rather than treating all missingness as equivalent.

## 3.2 Generate necessary variables

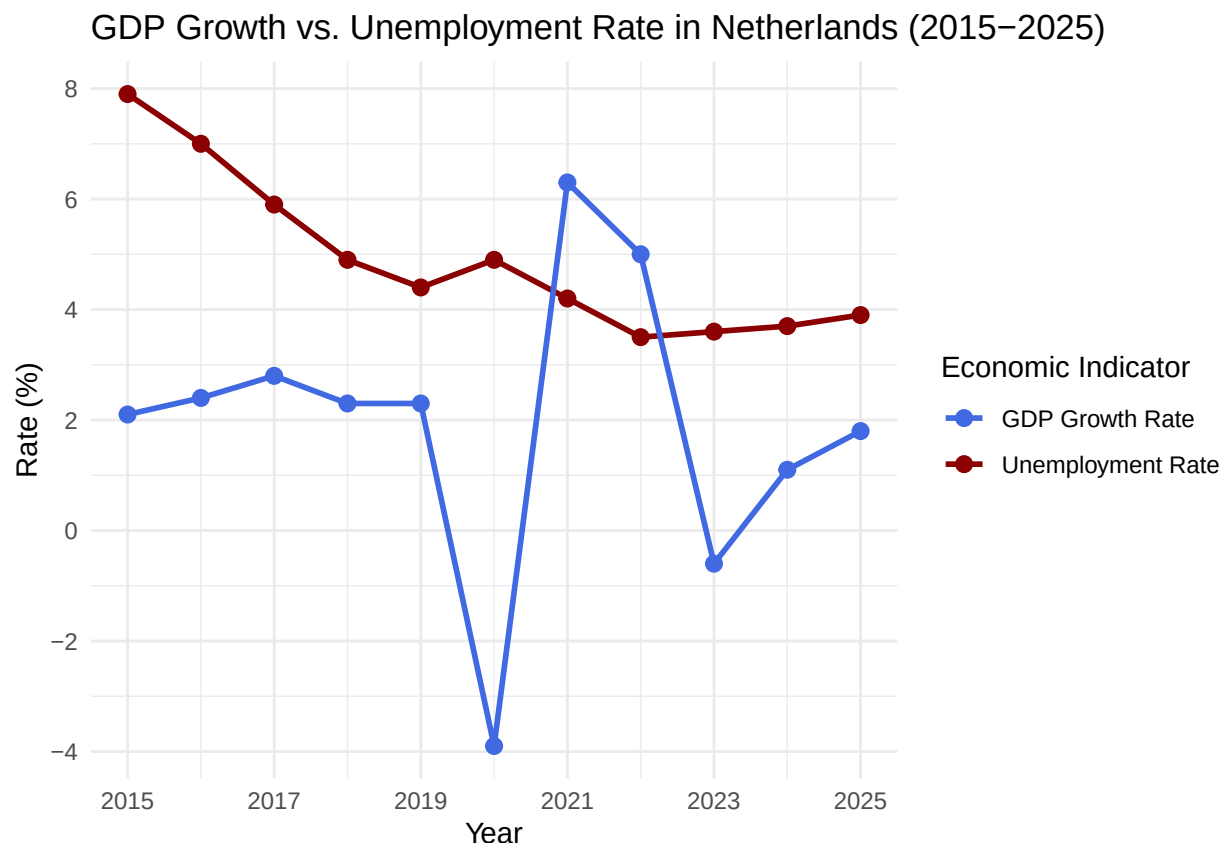
Variable 1: Youth Vulnerability Ratio

This variable divides the youth unemployment rate by the total unemployment rate (`Youth_Unemp_Rate / Unemp_Rate`), created in the pipeline at the top. It shows how much more exposed young workers are than the workforce as a whole.

Variable 2: Annual change in unemployment (percentage points)

We construct the year-on-year change in the unemployment rate (`Ppt_Change`) to capture labour-market dynamics rather than levels. It is created in the event-analysis chunk in 3.6 using `Unemp_Rate - lag(Unemp_Rate)`, and is used to quantify how sharply unemployment moved around the 2020 shock.

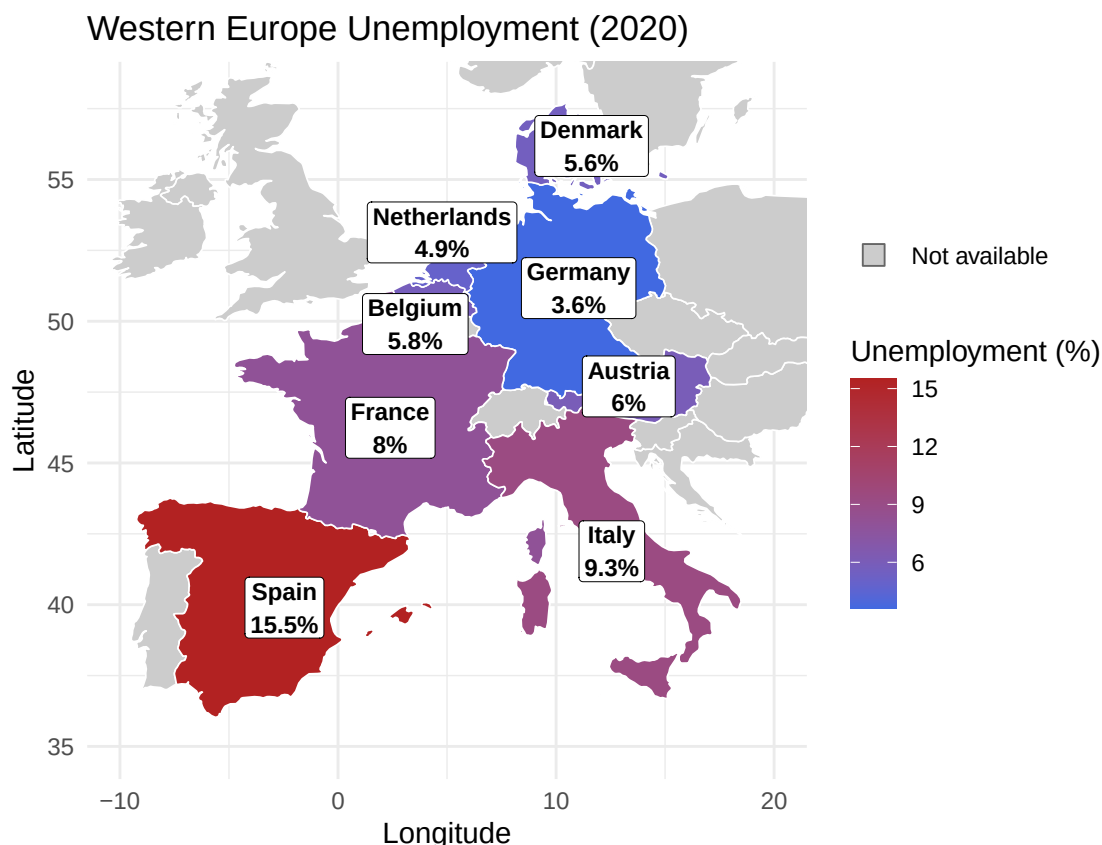
## 3.3 Visualize temporal variation



The two series show a clear inverse trend over most of the period: unemployment declines steadily from 2015 while GDP growth fluctuates around a positive average, consistent with Okun’s Law. The exception is 2020, when GDP collapses sharply but unemployment continues falling rather than spiking, breaking the otherwise visible inverse pattern.

### 3.4 Visualize spatial variation

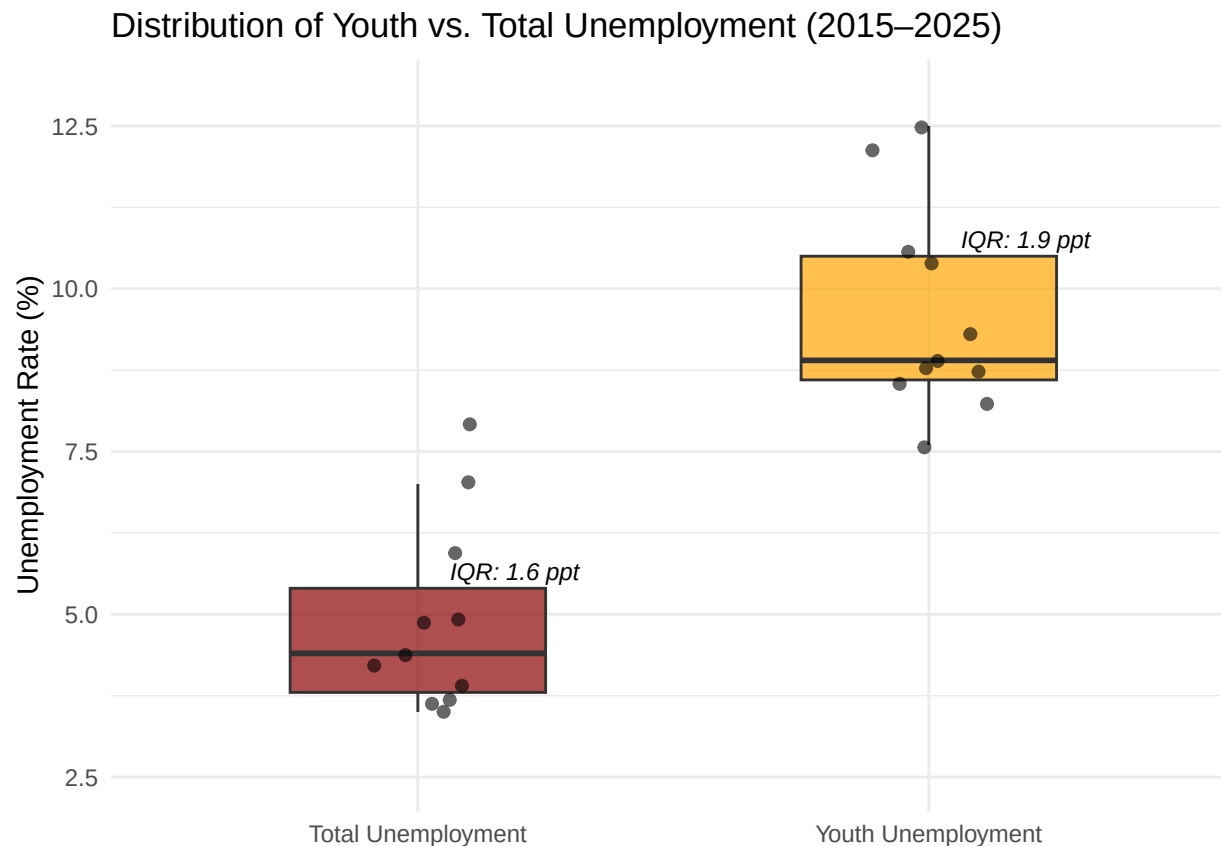
**Background:** all European countries, drawn in neutral grey



The map shows wide spatial variation in 2020 unemployment across Western Europe. The Netherlands (4.9%) and Germany (3.6%) sit at the low end, while Spain reaches 15.5% and Italy 9.3%. This clear north-south gradient places the Dutch labour market in context as relatively resilient, and motivates focusing the time-series analysis on the Netherlands.

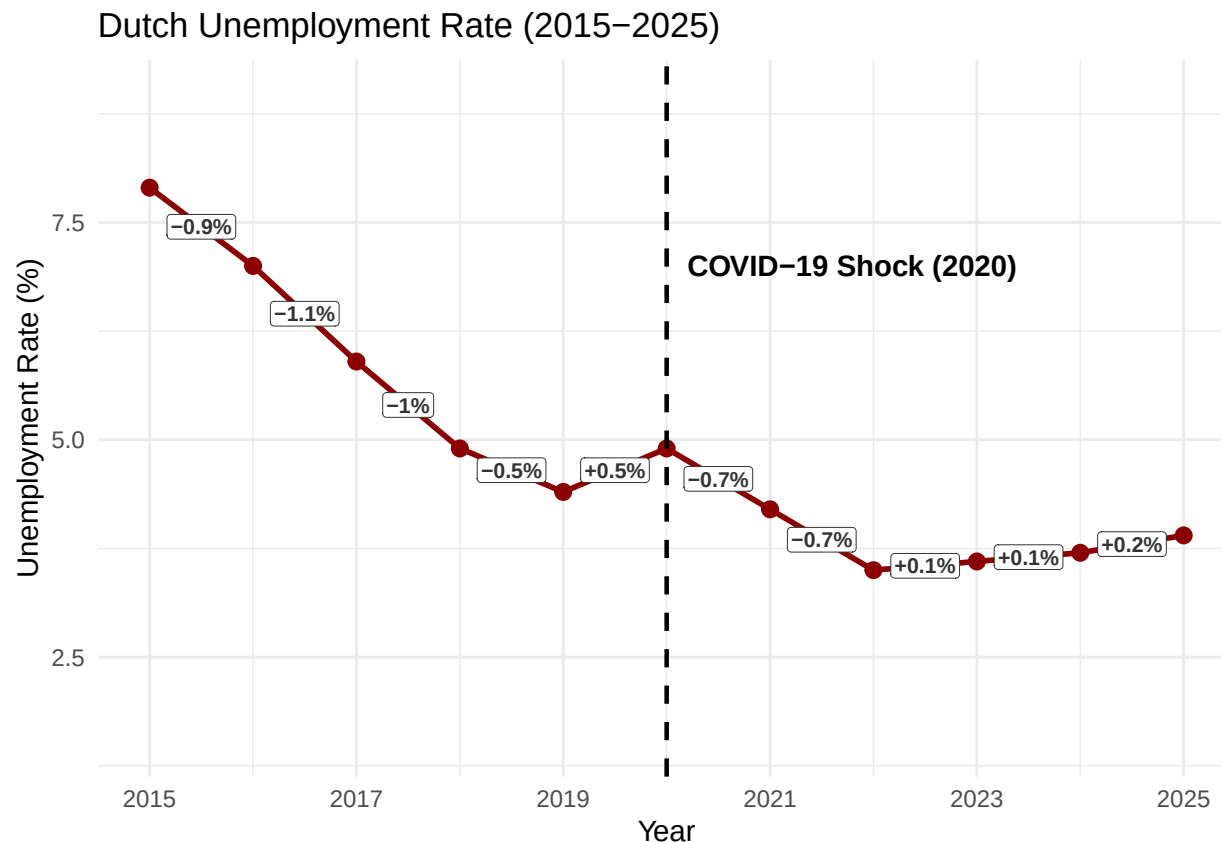
Two anomalies stand out against the broad north-south gradient. France (8.0%) sits well above its northern neighbours despite a comparable economy, and Spain's 15.5% is more than four times the German rate — a dispersion too large to be cyclical alone, pointing to structural labour-market differences. Our mapping choices also carry limits worth stating. We show only eight countries, chosen for Western-European comparability, which excludes high-unemployment Southern and Eastern states and so understates the EU-wide spread. We map a single year (2020), which is both the COVID shock year and our spatial snapshot; a different year would shift the gradient, so the map should be read as one cross-section, not a structural ranking. A choropleth filled by country rather than labelled points would also reduce the visual weight given to geographically large but low-population countries.

### 3.5 Visualize sub-population variation



Youth unemployment (ages 15–24) sits consistently far above the total rate across the whole period. The two distributions barely overlap. The youth box has a median of roughly 8.9% against about 4.4% for the total rate, so even the best youth years are worse than typical years for the workforce as a whole. The gap is not only in level but in spread: youth unemployment has a wider interquartile range (1.9 ppt vs 1.6 ppt) and longer whiskers, meaning it is more volatile year-to-year and reacts more sharply to downturns like the 2020 shock. This pattern of a higher floor and greater dispersion is what motivates our Youth Vulnerability Ratio. Young workers absorb a disproportionate and more variable share of labour-market shocks, and a single summary number understates how differently the two groups experience the same economy.

### 3.6 Event Analysis



The labelled changes show unemployment fell steadily before 2020 (around -1 ppt per year), then rose by only +0.5 ppt in the COVID-19 year. After this single-year interruption, the pre-2020 downward trend resumes, with unemployment falling again through 2022 before levelling off around 3.5-4% by 2025. This increase is remarkably small given the sharp GDP contraction that year, most likely because the Dutch emergency wage-subsidy scheme (NOW) kept workers formally employed, temporarily weakening the usual Okun's Law relationship.

## Part 4 - Discussion

### 4.1 Discuss your findings

Our analysis points to four findings. First, Dutch unemployment fell steadily from roughly 8% in 2015 to below 4% by 2025, interrupted only by a small rise around the 2020 COVID-19 shock. Second, GDP growth and unemployment broadly move in opposite directions, consistent with Okun's Law, but 2020 is a clear outlier: GDP fell sharply while unemployment barely moved, most likely because of the Dutch wage-subsidy scheme. Third, youth unemployment is consistently two to three times the total rate and moves more sharply with the cycle, as captured by our Youth Vulnerability Ratio. Fourth, the 2020 spatial comparison shows the Netherlands among the most resilient labour markets in Western Europe.

These results should be read with caution, and several limitations bound them. First, we document associations, not causal effects: GDP and unemployment are jointly shaped by labour-market policy, demographic change, and monetary conditions, and 2020 shows clearly how a raw correlation can mislead when a wage-subsidy scheme breaks the usual link. Second, the series is short (eleven annual observations), which makes

any estimated Okun coefficient fragile and rules out distinguishing a genuine structural break from noise. Third, the time-series analysis covers only the Netherlands, so findings need not generalise to countries with different labour-market institutions. Fourth, the spatial comparison rests on a single year and eight countries, limiting how far the cross-section can be read as structural. Finally, mixing a survey-based unemployment series with an administrative GDP series introduces definitional mismatch that we cannot fully net out.

## Part 5 - Reproducibility

### 5.1 Github repository link

[https://github.com/jackievahtera/PFE\\_Group\\_Project](https://github.com/jackievahtera/PFE_Group_Project)

### 5.2 How to reproduce this report

The repository contains this report (`betaversion.Rmd`), a `data/` folder with the three Eurostat CSV extracts (`raw_gdp.csv`, `raw_unemployment_sex.csv`, `raw_unemployment_youth.csv`), the bibliography (`references.bib`), and a `README.md` with a project overview.

To reproduce the report: (1) clone the repository, (2) open `ProgrForEconom.Rproj` in RStudio, (3) make sure the three CSVs are in the `data/` folder, (4) install the required packages (`tidyverse`, `maps`), and (5) knit `betaversion.Rmd` to PDF. Knitting re-reads the raw data and regenerates every figure and statistic in the document. All file paths in the `.Rmd` are relative to the project root, so the project runs on any machine without changes.

### 5.3 Reference list

Eurostat. (2025). Real GDP growth rate; Total unemployment rate; Youth unemployment rate [Data sets, tables `tec00115`, `tesem120`, `tipslm80`]. <https://ec.europa.eu/eurostat>

Eurofound. (2017). *Long-term unemployment*. Publications Office of the European Union.

Okun, A. M. (1962). Potential GNP: Its measurement and significance. *Proceedings of the Business and Economic Statistics Section*, 98–104.